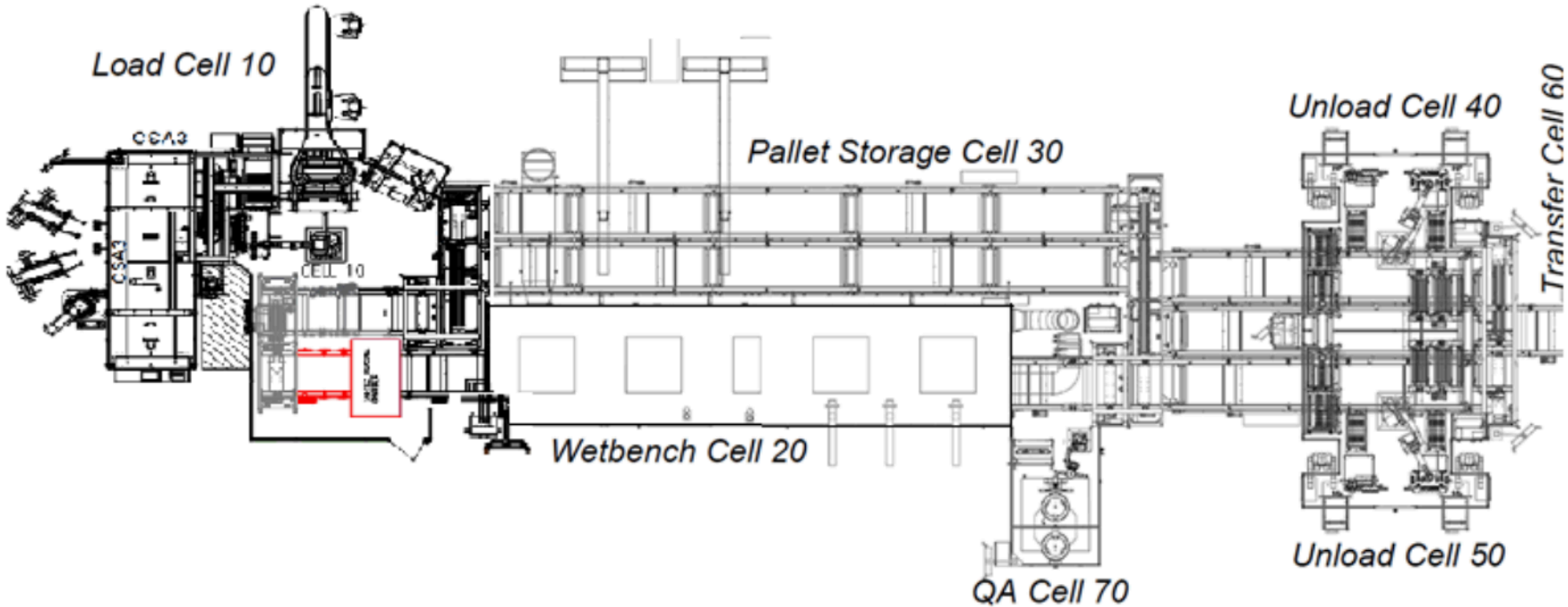


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VAPRO COATING LINE 5 (VCL5) – CELL 9

The purpose of this Manufacturing Instruction is to outline the procedures for operating the VaPro Coating Line 5. This is a three (3) person operation. VCL5 machine takes catheters, loads them onto pallets, applies a top coat, cures the top coat and unloads them into totes.





EMERGENCY STOP

In the event of an emergency push this button to STOP the machine.

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INTRODUCTION

Suppliers		Operation		Customers
Manual Loading of Catheters		VaPro Coating Line 5		Puncher Sleever

Responsibilities:

Team Leader and Training Coordinators: Manufacturing Instruction Implementation.
Operators: Adherence to the Manufacturing Instruction.

Learning Objectives:

At the end of this training activity the Trainee will be able to:

- Operate the VaPro Coating Line 5 in conjunction with the Work order in a safe and efficient manner.
- Operate the machine to produce a minimum target set by the Team Leader

Training Method:	Equipment:
Training Method will involve: • Job Talk • Practical Demonstration • Stamina Development	Listed below are the equipment requirements necessary for this operation: • VaPro Coating Line 5 • Isopropanol 70% • Gloves • Lint free wipes

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MATERIALS FOR VAPRO COATING LINE 5 (VLC5)

Catheters Subs

Supplied in tote boxes/bags.
Each tote box/bag has white WIP Ticket attached which contains the relevant details i.e. part number, machine number, Lot number, the date, shift and the quantity made.

Note: Materials status is clearly marked by a red “QA acceptance” Stamp. Only materials with this stamp may be used.

Catheter Subs are in different widths and lengths. VCL5 can produce Leapfrog Standard with Tip product.

WIP Ticket colours and paper

Leapfrog Standard With Tip	Yellow
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MATERIALS FOR VAPRO COATING LINE 5 (VLC5)

Ethanol 99% This is stored in an IBC container and is located in the Chemstore. Methanol is used to control the viscosity of the top coat.

The Ethanol is fed via the '**FMS**' (Fluid Management System) direct into the **VCL** machine.



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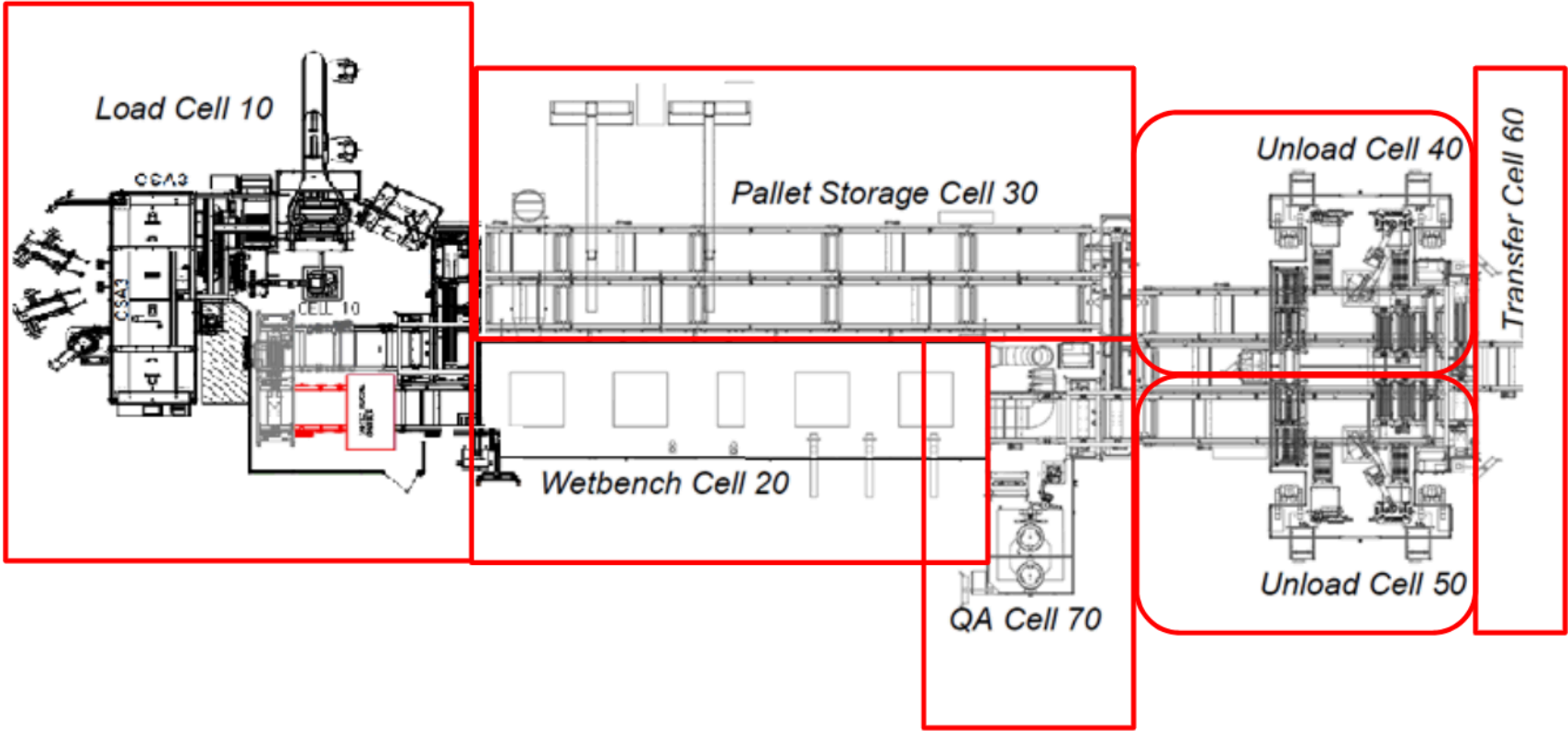
MATERIALS FOR VAPRO COATING LINE 5 (VLC5)

LCC – LCC is stored in Chem Store 2 and is fed via the ‘FMS’ (Fluid Management System) direct into the VCL Dip station.



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MACHINE OVERVIEW



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MACHINE OVERVIEW

Catheter Load Station (Cell 10)

Catheters are Loaded through the manual load station on VCL5. Ensure the catheters from the CSA have been produced 2 hours prior to loading into the VCL. The Manual Load station has 10 pallets that are carried by a conveyor handfed by operators. These pallets are fed into the Cell 10 robot where the catheters are loaded onto the pallets.

There are 24 catheters per fork, and 7 forks per pallet.

Once picked, the forks are held in front of the vision system where it will inspect the funnel colour and ensure funnels are fully attached on to the fork. If the vision system picks up a failure, it places the fork into the fork change and picks up a new fork. The rejected fork can then be manually stripped for catheter recovery.

The MRP selection at Cell 10 can only be produced when a complete line clearance is completed and the new MRP selection has a final destination (PS or WIP station).

The choice of what part to run comes from the pallet ordering system at the end of the line. The HMI tells the operators what type of catheter needs to be run to match what is being ordered by the PS machines.



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MACHINE OVERVIEW

Wet Bench Station (Cell 20)

The VCL5 Wet Bench is performs four main functions

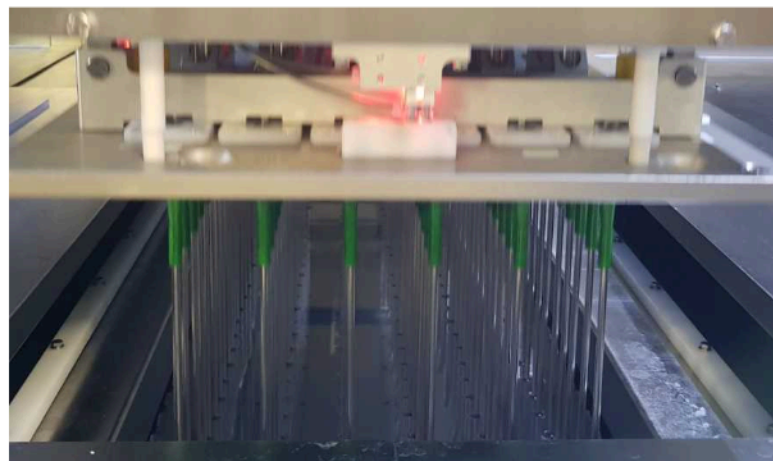
- Transferring of catheter pallets through the process
- Application of Topcoat solution
- UV curing of the Catheters

Step 1 - Catheters are received from the **Cell 10 infeed conveyor** and moved into the **Wetbench infeed Airlock**. Wetbench Gantry #1 then picks the pallet and transfers it to Park position 1 (HMI- Wash Tank 2)

Step 2 – Wetbench Gantry #2 picks the pallet from Park #1 and moves it to Park position #2 (HMI – Primer Cure) and held in the tank for a set period of time.

Step 3 – Wetbench Gantry #3 picks the pallet from Park #2 and moves the pallet to the Topcoat Dip Tank. Once the Topcoat dip process is complete. The pallet is then transferred into the flip station where the pallet is flipped 180Deg for 30 seconds.

Step 3 – The Catheter pallets are the picked by Gantry #4 to one of three **Topcoat UV cure** stations. When the topcoat cure process is complete, the processed catheter pallets are moved to the **Wetbench Outfeed Interlock**.



Catheters been loaded into tank

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MACHINE OVERVIEW

Process Flow of VCL5 Wetbench



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MACHINE OVERVIEW

Pallet Storage (Cell 30)

This cell has an elevator that takes pallets up from Cell 70 to a transfer which then places them on one of the two Buffer conveyors. This transfer makes the decision of where to send the pallet based on SKU, space, and PS availability. The cell has doors with locking switches. Access to the cell can be done through the HMI.

These are overhead conveyors that take full pallets from Cell 30 - Transfer 3 and deliver them to Cell 60 - Transfer 4, and they take empty pallets from Transfer 4 and deliver them down the middle back to Transfer 3 for delivery to the pallet storage conveyors



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MACHINE OVERVIEW

Unload Station to PS5 / 7 and PS6 / 8 (Cells 40 and 50)

Both Cell 40 and Cell 50 behave the same. Each cell feeds two PS machines simultaneously.

The pallet is brought to a robot on a conveyor and the robot picks the entire pallet and places it in one of the two pallet nests depending on part number and availability.

The robot picks a fork and brings it to the unload where the robot retracts, lifting the mandrels out of the catheters. The unload station then moves to the PS machines inline feed tracks and feeds catheters two at a time into tracks.

There is no manual unload station in VCL5. If the PS is not online, pallets are either unloaded to the other PS machine or to WIP (work in progress) depending on the part number.

On the occasion where a PS machine is stopped there is a WIP station that unloads product to totes. The WIP station consists of a cart and bay for empty totes and a cart and bay for full trays. The totes are deep enough to accommodate 168 catheters (a single pallet). The HMI is used to choose this option.



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MACHINE OVERVIEW

Transfer Station (Cell 60)

This station facilitates the transfer of pallets from Cell 30 (Transfer 3 and Buffers) to the PS machines and unload station. When pallets are empty in Cell 40 and 50 and unload they pass through Cell 60 which places them back to Cell 30 (Elevator 2, Transfer 3 and Buffers).

It is also the station where rejected pallets exit the VCL5 line and also where empty pallets are placed into Cell 60 to return back to Cell 30 Pallet Storage.

In order to remove or install pallets you need to pull back each pallet stop. Before placing pallet on track make sure to select correct mandrel length first. Place pallet in and lock on each pallet stop after pallet in correctly, making sure RFID tag is lined up with the RFID reader.



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MACHINE OVERVIEW

QA Station (Cell 70)

Immediately following the wet bench is Cell 70 which handles QA sampling and the rejecting of catheters from any pallet rejected in the wet bench.

The main purpose is to collect the dye and CoF samples at the desired rate. Typically the dye sample is generated by a random number from 1 to 168.

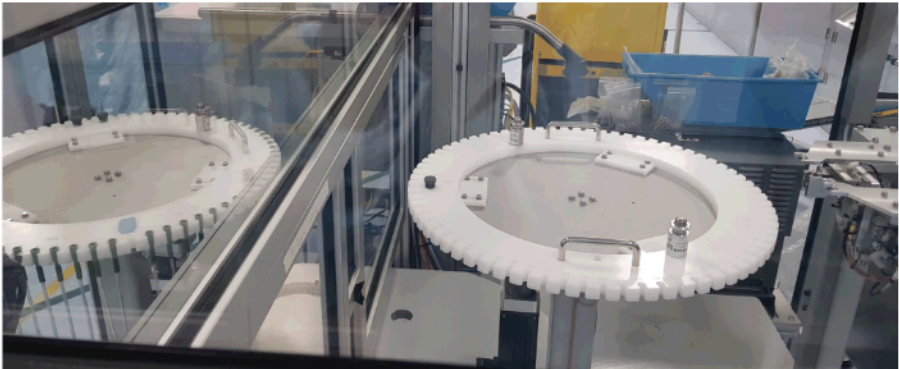
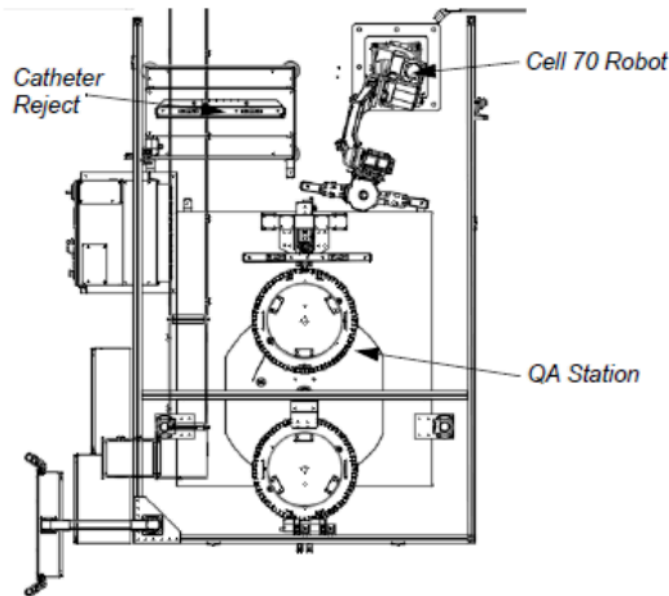
The QA station consists of two rings; each ring has 72 notches for sample catheters. The machine has two rings so one ring can be presented to the operator for removal and an empty replaced.

There is a safety door on the outside to prevent operator entrance while the ring is rotating.

The cell has one walk in door protected by a trap key system. It also has an interlocked door for protecting operators from the rotating dial.

Cell 70 contains a catheter reject station that can peel all 24 catheters off a fork and let them fall into a reject bin below the station. If a pallet fails for any reason in the Wetbench, all forks are peeled and returned to the pallet.

The quantity and reason for all rejected pallets are collected and displayed on the HMI.



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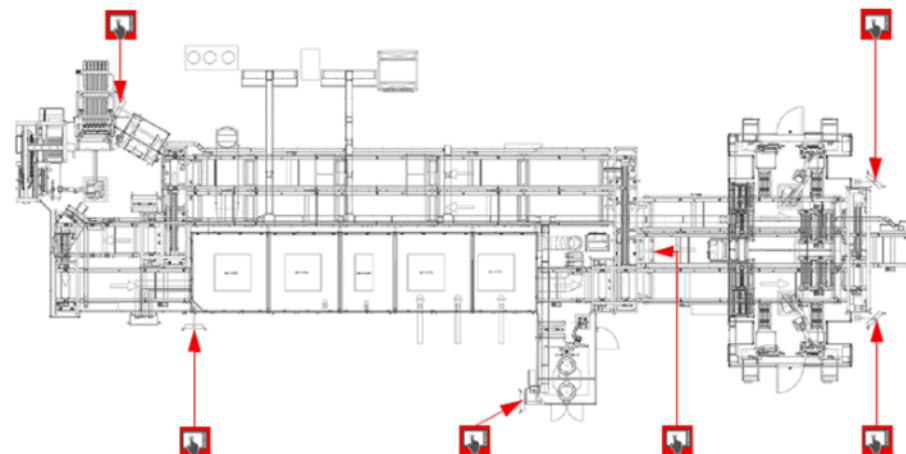
HMI OVERVIEW

VCL5 has six HMI screens set up in various areas around the VCL. Each HMI can control the general area it is located at but not the entire VCL. However, each HMI can view the status screen of other areas on the line.

However, there are some HMI's which allow control to other HMI's. The HMI's at the QA station can access the Wetbench HMI. The HMI's at the punching and sleeving area can also access the pallet storage HMI's and vice versa.

Pallet storage can access cell 40 and 50
70 can access 20

To allow switch control between the HMI's there is a button which requests control and notifies the other HMI that control is about to be taken. The white light indicates that you have control of the screen.



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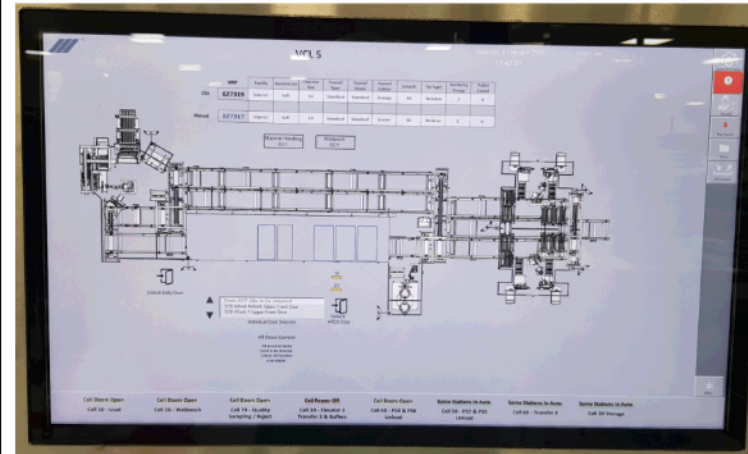
HMI OVERVIEW

Each of the 6 HMI's can view the same information. By clicking on Line Overview on the top right hand corner of the screen

Touching the Line Overview on the top right hand corner will give a general overview on the entire VCL process.

To go into each individual cell / station touch the appropriate cell / station. Alternatively you can click on the image to go into the individual cell / station.

To view a station fault the cell will be highlighted in red to indicate the fault. The fault is to be cleared on the HMI that corresponds to the station. For example Cell 50 – PS7 & PS5 cannot be cleared on the Cell 10 – Load HMI.



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END OF SHIFT

- At the end of each shift, the operator takes numbers from the HMI screen and reset the HMI screen. The operator must fill in the relevant details to complete the production/process/parameter sheets and place them on the Group Leader's desk.
- At the end of the shift, the operator must ensure to report the product count and the amount of calculated waste from the screen into the Hand Held Terminal (HHT) (See how to report goods in the SAP – Training Guide for Production Operators 75GEN00060_Att6). The operator must also ensure to complete the relevant parameter checks on the parameter check sheet and place it on the Group Leader's desk.
- The machine must be kept neat and tidy at all times.

Note: All surfaces of the VaPro Coating Line Machine and associated work benches must be cleaned with 70% IPA 3 times per shift as per 75CLR00001. A record of this cleaning must be recorded in the Production Sheet 75CLR00697_Att 2

To ensure that cleaning is carried out in the correct manner please follow the guidelines below;

- always wear gloves to protect your skin before using IPA
- always spray IPA onto the wipe, not the surface being cleaned - this will prevent IPA being sprayed into the eyes
- always ensure that you use enough IPA to wet the surface being cleaned

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